

数学笔记

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Chapter 1

Homotopy Theory

All spaces under consideration are connected.

1.1 Mapping Cylinder and Mapping Path Space

In the homotopy theory, every map $f : A \rightarrow B$ can be viewed as an inclusion or a fibration up to homotopy.

Inclusion: The mapping cylinder of f is defined as

$$M_f = (A \times I) \sqcup B / (a, 1) \sim f(a).$$

We have an inclusion $j : A \hookrightarrow M_f$ defined by

$$\begin{aligned} j : A &\hookrightarrow M_f \\ a &\mapsto (a, 0), \end{aligned}$$

an inclusion $i : B \hookrightarrow M_f$ defined by

$$\begin{aligned} i : B &\hookrightarrow M_f \\ b &\mapsto b, \end{aligned}$$

and a retraction $r : M_f \rightarrow B$ defined by

$$\begin{aligned} r : M_f &\rightarrow B \\ (a, t) &\mapsto f(a); \\ b &\mapsto b. \end{aligned}$$

Note that $r \circ i = \text{id}_B$ and $i \circ r \simeq \text{id}_{M_f}$ through the homotopy

$$H : M_f \times I \rightarrow M_f$$

$$\begin{aligned} ((a, t), s) &\mapsto (a, (1 - s)t + s); \\ (b, s) &\mapsto b. \end{aligned}$$

Geometrically, the homotopy H can be viewed as the process of continuously collapsing the cylinder $A \times I$ in M_f down onto the image of f in B .

Since $r \circ j = f$, we have a commutative diagram

$$\begin{array}{ccc} & M_f & \\ j \nearrow & & \searrow \simeq \\ A & \xrightarrow{f} & B \end{array}$$

in this sense, f is homotopic to the inclusion j .

Fibration: The mapping path space of f is defined as

$$P_f = \left\{ (a, \gamma) \in A \times \mathcal{P}(B) \mid \gamma(1) = f(a) \right\}.$$

where

$$\mathcal{P}(B) = \{ \gamma : I \rightarrow B \}$$

is the path space of B having arbitrary starting point and ending point.

We have a fibration $p : P_f \rightarrow B$ defined by

$$\begin{aligned} p : P_f &\rightarrow B \\ (a, \gamma) &\mapsto \gamma(0). \end{aligned}$$

The fiber of p over a point $b \in B$ is given by

$$\Omega_b^A := p^{-1}(b) = \left\{ (a, \gamma) \in A \times \mathcal{P}(B) \mid \begin{array}{l} \gamma(0) = b \\ \gamma(1) = f(a) \end{array} \right\}.$$

We have an inclusion $j : A \hookrightarrow P_f$ defined by

$$\begin{aligned} i : A &\hookrightarrow P_f \\ a &\mapsto (a, c_{f(a)}), \end{aligned}$$

where $c_{f(a)}$ is the constant path at $f(a)$, and a retraction $r : P_f \rightarrow A$ defined by

$$\begin{aligned} r : P_f &\rightarrow A \\ (a, \gamma) &\mapsto a. \end{aligned}$$

Note that $r \circ i = \text{id}_A$ and $i \circ r \simeq \text{id}_{P_f}$ through the homotopy

$$\begin{aligned} H : P_f \times I &\rightarrow P_f \\ ((a, \gamma), s) &\mapsto (a, \tilde{\gamma}_s). \end{aligned}$$

where $\tilde{\gamma}_s(t) = \gamma(s + (1-s)t)$ for $t \in I$ is a path in B that starts at $\gamma(s)$ and ends at $\gamma(1)$. Geometrically, the homotopy H can be viewed as the process of continuously collapsing the path γ in P_f down onto the constant path $c_{f(a)}$.

Since $r \circ j = f$, we have a commutative diagram

$$\begin{array}{ccc} & P_f & \\ \nearrow \simeq & & \searrow p \\ A & \xrightarrow{f} & B \end{array}$$

in this sense, f is homotopic to the fibration p .

注. The mapping cylinder can be constructed as a pushout of the diagram

$$\begin{array}{ccc} A & \xhookrightarrow{\iota_0} & A \times I \\ f \downarrow & \lrcorner & \downarrow \\ B & \longrightarrow & M_f \end{array}$$

where $\iota_0 : A \hookrightarrow A \times I$ is the inclusion map defined by $\iota_0(a) = (a, 0)$.

The mapping path space can be constructed as a pullback of the diagram

$$\begin{array}{ccc} P_f & \longrightarrow & A \\ \downarrow & \lrcorner & \downarrow f \\ \mathcal{P}(B) & \xrightarrow{\text{ev}} & B \end{array}$$

where $\text{ev} : \mathcal{P}(B) \rightarrow B$ is the evaluation map defined by $\text{ev}(\gamma) = \gamma(0)$.

1.1.1 Postnikov tower and Whitehead tower

We know that the homotopy type of a space X can be captured by its homotopy groups $\pi_q = \pi_q(X)$. In homotopy theory, the simplest spaces are the Eilenberg-MacLane spaces $K(A, n)$. So there is a natural question:

Can we decompose X as a "product" of a sequence of Eilenberg-MacLane spaces?

The directly product of Eilenberg-MacLane spaces is not enough to capture the homotopy type of X in general, i.e.,

$$X \not\cong \prod_q K(\pi_q(X), q).$$

So we need to introduce a more general construction, the Postnikov tower, and the Whitehead tower.

定义 1.1.1 (Postnikov tower). *Given a CW complex X and an integer n , the Postnikov tower of X is a sequence of fibrations*

$$\begin{array}{ccccc} K(\pi_n, n) & \longrightarrow & Y_n & & \\ & & \downarrow & \nearrow & \\ & & \vdots & & \\ & & \downarrow & \nearrow & \\ K(\pi_2, 2) & \longrightarrow & Y_2 & & \\ & & \downarrow & \nearrow & \\ K(\pi_1, 1) & \xlongequal{\quad} & Y_1 & \longleftarrow & X \end{array}$$

satisfying the following properties:

•

$$\pi_k(Y_q) = \begin{cases} 0 & , k > q; \\ \pi_k(X) & , k \leq q. \end{cases}$$

- the fibration $Y_q \rightarrow Y_{q-1}$ has fiber $K(\pi_q, q)$ and commutes with the maps $X \rightarrow Y_q$;
- the map $X \rightarrow Y_q$ induces isomorphisms

$$\pi_k(X) \xrightarrow{\cong} \pi_k(Y_q), \quad k \leq q.$$

Construction: The space Y_q is the space obtained from X by truncating its homotopy groups above degree q .

Step 1: Construct Y_n from X

To construct Y_n , we have to kill the homotopy groups $\pi_k(X)$ for $k > n$. The method is attaching cells. To be more precise, if $\alpha : S^{n+1} \rightarrow X$ is a generator of $\pi_{n+1}(X)$, we can attach a cell e^{n+2} to X , i.e.,

$$X \cup_{\alpha} e^{n+2} = X \sqcup e^{n+2} / x \sim \alpha(x)$$

to kill the generator α in $\pi_{n+1}(X)$, Repeat this process for all generators of $\pi_{n+1}(X)$, we get X_1 . Note that this process does not change the homotopy groups $\pi_k(X)$ for $k \leq n$. Then we can repeat this process for X_1 to get Y_n .

$$Y_n = X \cup_{\alpha} e^{n+2} \cup \dots$$

we have a natural inclusion $X \hookrightarrow Y_n$.

Step 2: Inductively construct Y_{q-1} from Y_q

To construct Y_{q-1} from Y_q , we have to kill $\pi_q(Y_q)$. The method is similar to the previous step, i.e., attaching cells.

$$Y_{q-1} = Y_q \cup e^{q+1} \cup \dots$$

therefore we have a natural inclusion $Y_q \hookrightarrow Y_{q-1}$. It can be viewed as a fibration

$$\begin{array}{ccc} F_q & \longrightarrow & Y_q \\ & & \downarrow \\ & & Y_{q-1} \end{array}$$

Step 3: Check F_q is $K(\pi_q, q)$

The fibration induces a long exact sequence of homotopy groups

$$\dots \rightarrow \pi_{k+1}(Y_q) \rightarrow \pi_{k+1}(Y_{q-1}) \rightarrow \pi_k(F_q) \rightarrow \pi_k(Y_q) \rightarrow \pi_k(Y_{q-1}) \rightarrow \dots$$

- $k \geq q + 1$: $\pi_{k+1}(Y_{q-1}) = \pi_k(Y_q) = 0$, so $\pi_k(F_q) = 0$.
- $k = q$: $\pi_{q+1}(Y_{q-1}) = \pi_q(Y_{q-1}) = 0$, so $\pi_q(F_q) = \pi_q(Y_q) = \pi_q(X)$.
- $k = q - 1$: $\pi_q(Y_{q-1}) = 0$, $\pi_{q-1}(Y_q) \cong \pi_{q-1}(X)$, so $\pi_{q-1}(F_q) = 0$.
- $k \leq q - 2$: $\pi_{k+1}(Y_q) \cong \pi_{k+1}(Y_{q-1})$, so $\pi_k(F_q) = 0$.

Therefore, we have

$$\pi_k(F_q) = \begin{cases} \pi_q(X) & , k = q; \\ 0 & , k \neq q. \end{cases}$$

So $F_q \simeq K(\pi_q, q)$. □

定义 1.1.2 (Whitehead tower). *Given a CW complex X , the Whitehead tower of X is a sequence of fibrations*

$$\begin{array}{ccc}
& & \vdots \\
& & \downarrow \\
K(\pi_n, n-1) & \longrightarrow & X_n \\
& & \downarrow \\
K(\pi_{n-1}, n-2) & \longrightarrow & X_{n-1} \\
& & \downarrow \\
& & \vdots \\
& & \downarrow \\
K(\pi_1, 0) & \longrightarrow & X_1 \\
& & \downarrow \\
& & X
\end{array}$$

satisfying the following properties:

•

$$\pi_k(X_n) = \begin{cases} 0 & , k \leq n; \\ \pi_k(X) & , k > n. \end{cases}$$

• the fibration $X_n \rightarrow X_{n-1}$ has fiber $K(\pi_n, n-1)$.

Construction: The space X_1 is the universal cover of X in the homotopy sense. The space X_n is the generalization of X_1 to higher dimensions.

Step 1: Construct X_1 from X

To construct X_1 , we have to kill $\pi_1(X)$. First kill the higher homotopy groups of X to get

$$Y_1 = X \cup e^3 \cup \cdots = K(\pi_1(X), 1)$$

X is naturally included in Y_1 . We define

$$X_1 = \Omega_*^X = \left\{ (x, \gamma) \in X \times \mathcal{P}(Y_1) \left| \begin{array}{l} \gamma(0) = * \\ \gamma(1) = x \end{array} \right. \right\}.$$

The projection $p : X_1 \rightarrow X$ is a fibration with fiber

$$p^{-1}(*) = \Omega Y_1 = K(\pi_1, 0).$$

We can show that

$$\pi_k(X_1) = \begin{cases} \pi_k(X) & , k \geq 2; \\ 0 & , k \leq 1. \end{cases}$$

So X_1 is the universal cover of X in the homotopy sense.

Step 2: Inductively construct X_n from X_{n-1}

Firstly, kill the higher homotopy groups of X_{n-1} to get

$$Y_n = X_{n-1} \cup e^{n+2} \cup \dots = K(\pi_n, n).$$

Then we can repeat the process for X_n to get

$$X_n = \Omega_*^{X_{n-1}}$$

The projection $p : X_n \rightarrow X_{n-1}$ gives a fibration

$$\begin{array}{ccc} K(\pi_n, n-1) = \Omega Y_n & \longrightarrow & X_n \\ & & \downarrow \\ & & X_{n-1} \end{array}$$

Step 3: Check $\pi_k(X_n)$

The fibration induces a long exact sequence of homotopy groups

$$\dots \rightarrow \pi_{k+1}(X_{n-1}) \xrightarrow{\delta} \pi_k(\Omega Y_n) \rightarrow \pi_k(X_n) \rightarrow \pi_k(X_{n-1}) \xrightarrow{\delta} \pi_{k-1}(\Omega Y_n) \rightarrow \dots$$

- $k \geq n+1$: $\pi_k(\Omega Y_n) = \pi_{k-1}(\Omega Y_n) = 0$. So $\pi_k(X_n) \cong \pi_k(X_{n-1}) = \pi_k(X)$.
- $k = n, n-1$:

$$\pi_n(\Omega Y_n) \rightarrow \pi_n(X_n) \rightarrow \pi_n(X_{n-1}) \xrightarrow{\delta} \pi_{n-1}(\Omega Y_n) \rightarrow \pi_{n-1}(X_n) \rightarrow \pi_{n-1}(X_{n-1})$$

The connecting homomorphism $\delta : \pi_n(X_{n-1}) \rightarrow \pi_{n-1}(\Omega Y_n)$ is an isomorphism, $\pi_n(\Omega Y_n) = \pi_{n+1}(Y_n) = 0$, $\pi_{n-1}(X_{n-1}) = 0$. So $\pi_n(X_n) = 0$.

- $k \leq n-2$: $\pi_k(\Omega Y_n) = \pi_k(X_{n-1}) = 0$. So $\pi_k(X_n) = 0$.

Therefore, we have

$$\pi_k(X_n) = \begin{cases} \pi_k(X) & , k \geq n+1; \\ 0 & , k \leq n. \end{cases}$$

Why δ is an isomorphism?

The connecting homomorphism δ is defined as follows:

$$\begin{aligned}\delta : \pi_n(X_{n-1}) &\rightarrow \pi_{n-1}(\Omega Y_n) \\ [\alpha] &\mapsto [\tilde{\alpha}],\end{aligned}$$

where $\alpha : (I^n, \partial I^n) \rightarrow (X_{n-1}, *)$ represents $[\alpha]$. $\tilde{\alpha} : (I^{n-1}, \partial I^{n-1}) \rightarrow (\Omega Y_n, c_*)$ is defined by

$$\tilde{\alpha}(s)(t) = \alpha(s, t), \quad (s, t) \in I^{n-1} \times I = I^n.$$

This is essentially treating the last coordinate of α as the time parameter of the path. Note that the connecting homomorphism

$$\begin{aligned}\delta' : \pi_n(Y_n) &\xrightarrow{\cong} \pi_{n-1}(\Omega Y_n) \\ [\alpha] &\mapsto [\tilde{\alpha}],\end{aligned}$$

is defined in the same way. δ is exactly the composition of two isomorphisms:

$$\begin{array}{ccccc} & & \delta & & \\ & \nearrow & & \searrow & \\ \pi_n(X_n) & \xrightarrow[\iota_*]{\cong} & \pi_n(Y_n) & \xrightarrow[\delta']{\cong} & \pi_{n-1}(\Omega Y_n) \end{array}$$

Therefore, δ is an isomorphism. □

注. *Postnikov tower* 将空间 X 从小空间一点一点拼装起来; *Whitehead tower* 则是将空间 X 从基本群开始逐步解开.

在 *Postnikov tower* 中, 每一项 Y_n 具有特殊的胞腔分解

$$Y_n = X \cup e^{n+2} \cup \dots$$

所以能得到 Y_n 的同调群信息.

在 *Whitehead tower* 中, 每一项 X_n 都是 n -连通空间, 所以能得到 X_n 的同调群信息.

1.1.2 The rational homotopy groups of spheres

The rational homotopy groups of spheres are given by the following theorem:

定理 1.1.3 (Sere theorem). *If n is odd, then*

$$\pi_q(S^n) \otimes \mathbb{Q} = \begin{cases} \mathbb{Q} & , q = 0, n; \\ 0 & , \text{otherwise.} \end{cases}$$

If n is even, then

$$\pi_q(S^n) \otimes \mathbb{Q} = \begin{cases} \mathbb{Q} & , q = 0, n, 2n-1; \\ 0 & , \text{otherwise.} \end{cases}$$

Before proving this theorem, we need two lemmas.

引理 1.1.4. *All $\pi_q(S^n)$ are finitely generated abelian groups.*

The proof of this lemma is based on Serre's $\mathcal{C}$ -class. We skip the details here.

引理 1.1.5.

- For any finite abelian group A , $H^*(K(A, n); \mathbb{Q}) = 0$.
- The cohomology ring $H^*(K(\mathbb{Z}, n); \mathbb{Q})$ is free on one generator of degree n , i.e.,

$$H^*(K(\mathbb{Z}, n); \mathbb{Q}) = \begin{cases} \Lambda(x), \dim x = n, & n \text{ is odd;} \\ \mathbb{Q}[x], \dim x = n, & n \text{ is even.} \end{cases}$$

证明. For $n = 1$, $K(\mathbb{Z}, 1) \simeq S^1$, so $H^*(K(\mathbb{Z}, 1); \mathbb{Q}) \cong \Lambda(x)$, $\dim x = 1$.

For $n = 2$, $K(\mathbb{Z}, 2) \simeq \mathbb{C}P^\infty$, so $H^*(K(\mathbb{Z}, 2); \mathbb{Q}) \cong \mathbb{Q}[x]$, $\dim x = 2$.

For $n \geq 3$, we use the fibration

$$\begin{array}{ccc} K(\mathbb{Z}, 2) & \longrightarrow & \mathcal{P}K(\mathbb{Z}, 3) \\ & & \downarrow \\ & & K(\mathbb{Z}, 3) \end{array}$$

Its associated Serre spectral sequence has E_2 -term

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6	$\vdots$	$\searrow$	$\cdot 3$	$\vdots$
5				
4	x^2	$\searrow$	$\cdot 2$	ax^2
$H^*(\mathbb{C}P^\infty)$ 3				
2	x	$\searrow$	$\simeq$	ax
1				
0	$\mathbb{Q}$	$\searrow$		a

$H^*(K(\mathbb{Z}, 3))$

To cancel x , we have to set $d_3x = a$ to be an isomorphism. The above differential map is given by product structure. Since we are working with rational coefficients, they are all isomorphisms. So we have $E_4 = E_\infty$. Therefore, we have

$$H^*(K(\mathbb{Z}, 3); \mathbb{Q}) = \Lambda(x), \dim x = 3.$$

For $n \geq 4$, we can use the fibration

$$\begin{array}{ccc} K(\mathbb{Z}, 3) & \longrightarrow & \mathcal{P}K(\mathbb{Z}, 4) \\ & & \downarrow \\ & & K(\mathbb{Z}, 4) \end{array}$$

Its associated Serre spectral sequence has E_2 -term

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$H^*(K(\mathbb{Z}, 3))$	$\begin{array}{c} 3 \\ 2 \\ 1 \\ 0 \end{array}$	$\begin{array}{ccccccc} x & & ax & & a^2x & & \cdots \\ & \searrow & & \searrow & & \searrow & \\ & & a & & a^2 & & \cdots \end{array}$
	$\mathbb{Q}$	
	$\begin{array}{cccccccccccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \end{array}$	$H^*(K(\mathbb{Z}, 4))$

Therefore, we have

$$H^*(K(\mathbb{Z}, 4); \mathbb{Q}) = \mathbb{Q}[x], \dim x = 4.$$

By induction, we can get the result for all $n \geq 5$. □

Now we can prove the Sere theorem. There are two ways to prove it, one is using the Postnikov tower, the other is using the Whitehead tower.

Proof of Sere theorem using Postnikov tower. Since S^n is n -connected, we have the Postnikov tower

$$\begin{array}{ccc} & & \vdots \\ & & \downarrow \\ K(\pi_{n+2}, n+2) & \longrightarrow & Y_{n+2} \\ & & \downarrow \\ K(\pi_{n+1}, n+1) & \longrightarrow & Y_{n+1} \\ & & \downarrow \\ & & K(\mathbb{Z}, n) \end{array}$$

By the construction of the Postnikov tower, we have

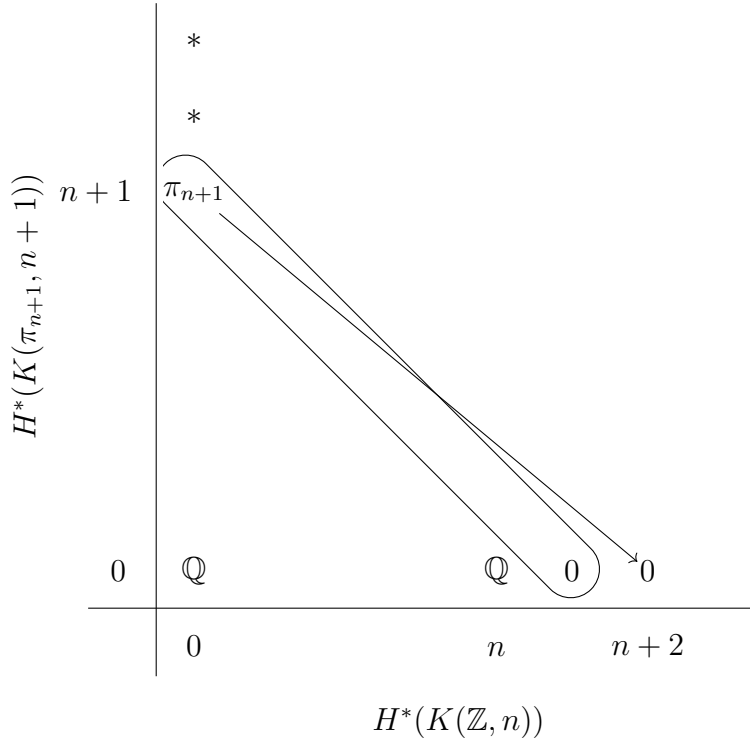
$$Y_q = S^n \cup e^{q+2} \cup \dots, \quad q \geq n+1.$$

Hence the cohomology groups of Y_q satisfies

$$H^q(Y_q; \mathbb{Q}) = H^{q+1}(Y_q; \mathbb{Q}) = 0.$$

If n is odd:

◦ $\dim = n+1$: The fibration $Y_{n+1} \rightarrow K(\mathbb{Z}, n)$ has spectral sequence



Since $K(\pi_{n+1}, n+1)$ is n -connected, by Hurewicz theorem, we have

$$H^{n+1}(K(\pi_{n+1}, n+1); \mathbb{Q}) = \pi_{n+1} \otimes \mathbb{Q}.$$

By the construction of the Postnikov tower,

$$Y_{n+1} = S^n \cup e^{n+2} \cup \dots$$

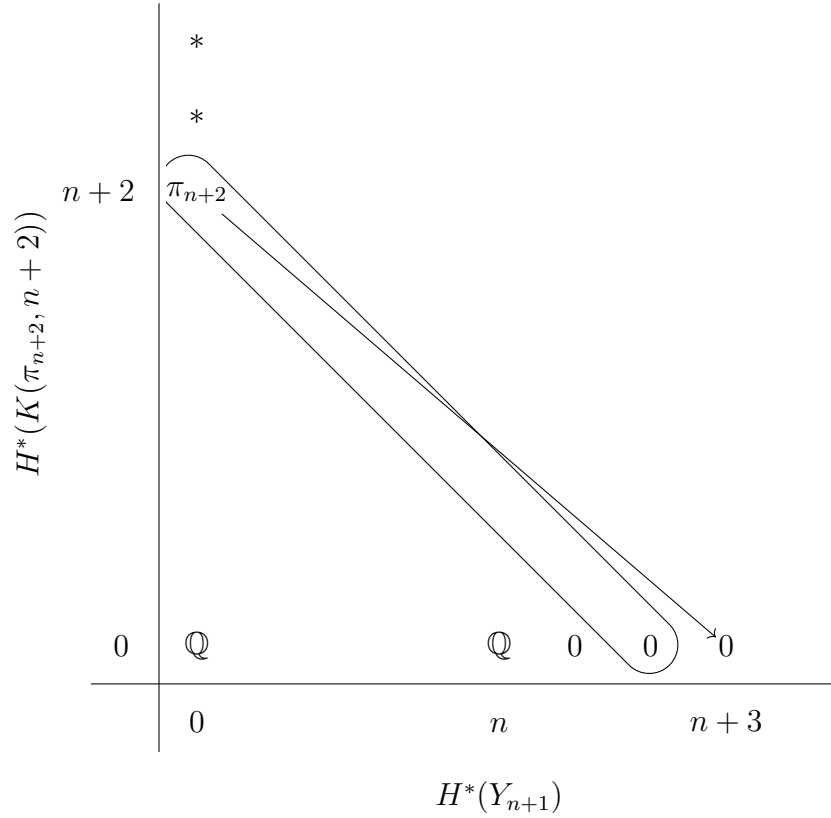
Therefore

$$H^{n+1}(Y_{n+1}; \mathbb{Q}) = 0 \implies \pi_{n+1} \otimes \mathbb{Q} = 0.$$

◦ $\dim = n + 2$: By lemma 1.1.5, $K(\pi_{n+1}, n + 1)$ has trivial rational cohomology, so

$$H^*(Y_{n+1}; \mathbb{Q}) \cong H^*(K(\mathbb{Z}, n); \mathbb{Q}).$$

The fibration $Y_{n+2} \rightarrow Y_{n+1}$ has spectral sequence



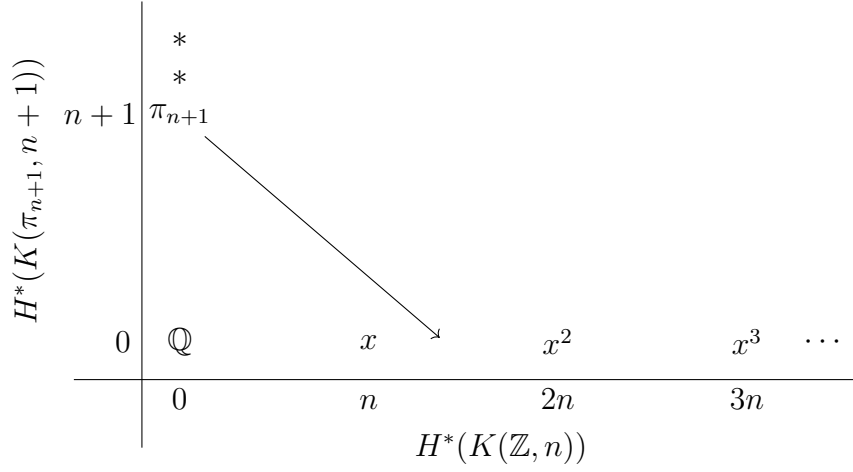
By the same argument as above, we have

$$H^{n+2}(Y_{n+2}; \mathbb{Q}) = 0 \implies \pi_{n+2} \otimes \mathbb{Q} = 0.$$

◦ $\dim \geq n + 3$: By induction.

If n is even:

◦ $\dim = n + 1$: The fibration $Y_{n+1} \rightarrow K(\mathbb{Z}, n)$ has spectral sequence



By the construction of the Postnikov tower,

$$Y_{n+1} = S^n \cup e^{n+2} \cup \dots$$

Therefore,

$$H^{n+1}(Y_{n+1}; \mathbb{Q}) = 0 \implies \pi_{n+1} \otimes \mathbb{Q} = 0.$$

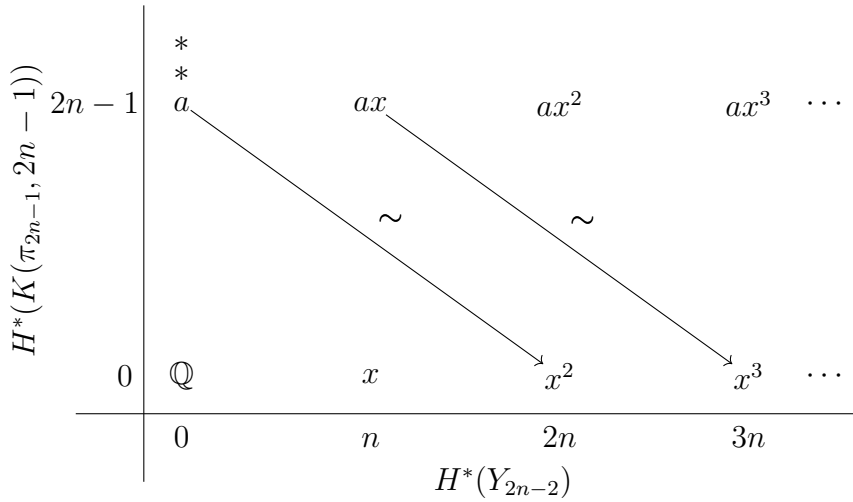
◦ $\dim = n + 2$: Since $K(\pi_{n+1}, n + 1)$ has trivial rational cohomology,

$$H^*(Y_{n+1}; \mathbb{Q}) \cong H^*(K(\mathbb{Z}, n); \mathbb{Q}) = \mathbb{Q}[x], \dim x = n.$$

The rest analysis is similar to $n + 1$ case.

.....

◦ $\dim = 2n - 1$: The fibration $Y_{2n-1} \rightarrow Y_{2n-2}$ has spectral sequence



By the same argument as above, we have

$$H^{2n-1}(Y_{2n-1}; \mathbb{Q}) = H^{2n}(Y_{2n-1}; \mathbb{Q}) = 0.$$

Therefore, $d_{2n} : \pi_{2n-1} \otimes \mathbb{Q} \rightarrow H^{2n}(Y_{2n-2}; \mathbb{Q})$ is an isomorphism. So

$$\pi_{2n-1} \otimes \mathbb{Q} = \mathbb{Q}.$$

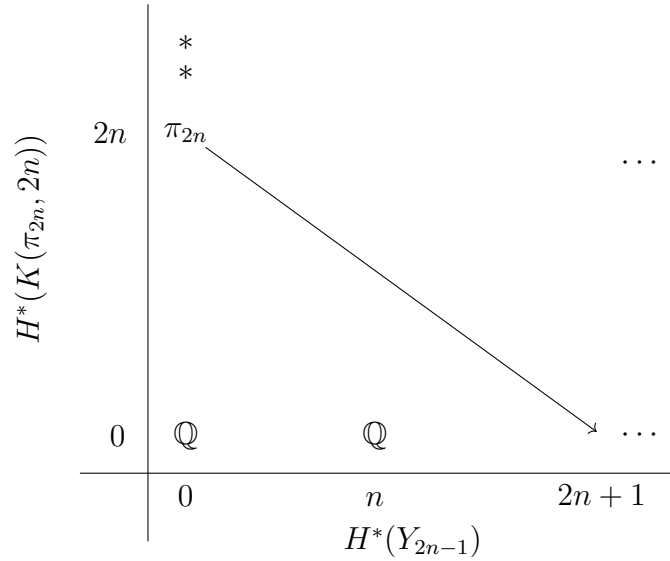
◦ $\dim = 2n$: By lemma 1.1.5,

$$H^*(K(\pi_{2n-1}, 2n-1); \mathbb{Q}) = \Lambda(x), \dim x = 2n-1.$$

Thus the unknown terms in the first column are all zero. So

$$H^*(Y_{2n-1}; \mathbb{Q}) = \begin{cases} \mathbb{Q}, & \dim = 0, n; \\ 0 & \text{otherwise.} \end{cases}$$

The fibration $Y_{2n} \rightarrow Y_{2n-1}$ has spectral sequence



By the same argument as above, we have

$$H^{2n}(Y_{2n}; \mathbb{Q}) = 0 \implies \pi_{2n} \otimes \mathbb{Q} = 0.$$

◦ $\dim = 2n+1$: Since $K(\pi_{2n}, 2n)$ has trivial rational cohomology

$$H^*(Y_{2n}; \mathbb{Q}) \cong H^*(Y_{2n-1}; \mathbb{Q}).$$

The rest analysis is similar to $2n$ case.

◦ $\dim \geq 2n+2$: By induction. □

Proof of Sere theorem using Whitehead tower. The Whitehead tower of S^n is given by the following fibration

$$\begin{array}{ccc} & & \vdots \\ & & \downarrow \\ K(\pi_{n+1}, n) & \longrightarrow & X_{n+1} \\ & & \downarrow \\ K(\mathbb{Z}, n-1) & \longrightarrow & X_n \\ & & \downarrow \\ & & S^n \end{array}$$

By the construction of the Whitehead tower, we have

$$\pi_k(X_q) = \begin{cases} \pi_k(S^n) & , k \geq q+1; \\ 0 & , k \leq q. \end{cases}$$

By Hurewicz theorem, the cohomology groups of X_n satisfies

$$H^q(X_q; \mathbb{Q}) = 0, \quad H^{q+1}(X_q; \mathbb{Q}) = \pi_{q+1}(S^n) \otimes \mathbb{Q}.$$

If n is odd:

◦ $\dim = n+1$: The fibration $X_{n+1} \rightarrow K(\mathbb{Z}, n)$ has spectral sequence

$$\begin{array}{c} \begin{array}{c} \vdots \\ 2(n-1) \\ n-1 \\ 0 \end{array} \begin{array}{c} x^2 \\ x \\ \mathbb{Q} \end{array} \begin{array}{c} \vdots \\ ax^2 \\ ax \\ a \end{array} \\ \begin{array}{c} 0 \\ n \end{array} \begin{array}{c} 0 \\ H^*(S^n) \end{array} \end{array}$$

$\begin{array}{ccc} & \searrow \sim & \\ & & \end{array}$
 $\begin{array}{ccc} & \searrow \sim & \\ & & \end{array}$

By lemma 1.1.5,

$$H^*(K(\mathbb{Z}, n-1); \mathbb{Q}) = \mathbb{Q}[x], \quad \dim x = n-1.$$

By the construction of the Whitehead tower, X_n is n -connected.

$$H^{n-1}(X_n; \mathbb{Q}) = H^n(X_n; \mathbb{Q}) = 0.$$

So $d_n : \pi_{n-1} \otimes \mathbb{Q} \rightarrow H^n(S^n; \mathbb{Q})$ must be an isomorphism. The other differentials are all isomorphisms by the product structure and rational coefficients. Therefore,

$$H^*(X_n; \mathbb{Q}) \text{ is trivial} \implies \pi_{n+1} \otimes \mathbb{Q} = H^{n+1}(X_n; \mathbb{Q}) = 0.$$

◦ $\dim = n+2$: In the fibration

$$\begin{array}{ccc} K(\pi_{n+1}, n) & \longrightarrow & X_{n+1} \\ & & \downarrow \\ & & X_n \end{array}$$

both $K(\pi_{n+1}, n)$ and X_n have trivial rational cohomology. Therefore,

$$H^*(X_{n+1}; \mathbb{Q}) \text{ is trivial} \implies \pi_{n+2} \otimes \mathbb{Q} = H^{n+2}(X_{n+1}; \mathbb{Q}) = 0.$$

◦ $\dim \geq n+3$: By induction.

If n is even:

◦ $\dim = n+1$: The fibration $X_{n+1} \rightarrow K(\mathbb{Z}, n-1)$ has spectral sequence

$$\begin{array}{ccc} & & \vdots \\ & & \mathbb{Q} \\ n-1 & & \mathbb{Q} \\ & \searrow \sim & \\ & & \mathbb{Q} \\ 0 & & \mathbb{Q} \\ & \nearrow & \\ & & \mathbb{Q} \\ & & n \\ & & H^*(S^n) \end{array}$$

Since X_n is n -connected, the differential map shown above must be an isomorphism. Therefore,

$$H^*(X_n; \mathbb{Q}) = \begin{cases} \mathbb{Q}, & \dim = 0, 2n-1; \\ 0, & \text{otherwise} \end{cases} \implies \pi_{n+1} \otimes \mathbb{Q} = H^{n+1}(X_n; \mathbb{Q}) = 0.$$

◦ $\dim = n+2$: In the fibration

$$\begin{array}{ccc}
K(\pi_{n+1}, n) & \longrightarrow & X_{n+1} \\
& & \downarrow \\
& & X_n
\end{array}$$

$K(\pi_{n+1}, n)$ has trivial rational cohomology. Therefore,

$$H^*(X_{n+1}; \mathbb{Q}) \cong H^*(X_n; \mathbb{Q}) \implies \pi_{n+2} \otimes \mathbb{Q} = H^{n+2}(X_{n+1}; \mathbb{Q}) = 0.$$

.....

◦ $\dim = 2n - 1$: The fibration $X_{2n-1} \rightarrow X_{2n-2}$ has spectral sequence

$$\begin{array}{ccc}
& \vdots & \vdots \\
4n-4 & x^2 & ax^2 \\
& \searrow \sim & \\
2n-2 & x & ax \\
& \searrow \sim & \\
0 & \mathbb{Q} & a \\
& \text{0} & 2n-1 \\
& H^*(X_{2n-2}) &
\end{array}$$

Since X_{2n-2} is $(2n-2)$ -connected, the differential maps shown above must be isomorphisms. Therefore,

$$H^*(X_{2n-1}; \mathbb{Q}) \text{ is trivial } \implies \pi_{2n-1} \otimes \mathbb{Q} = H^{2n-1}(X_{2n-1}; \mathbb{Q}) = 0.$$

◦ $\dim = 2n$: Since $K(\pi_{2n-1}, 2n-1)$ has trivial rational cohomology.

$$H^*(X_{2n}; \mathbb{Q}) \cong H^*(X_{2n-1}; \mathbb{Q}) \text{ is trivial } \implies \pi_{2n} \otimes \mathbb{Q} = 0.$$

◦ $\dim \geq 2n + 1$: By induction. □

1.2 Homotopy groups

定义 1.2.1. *omit.*

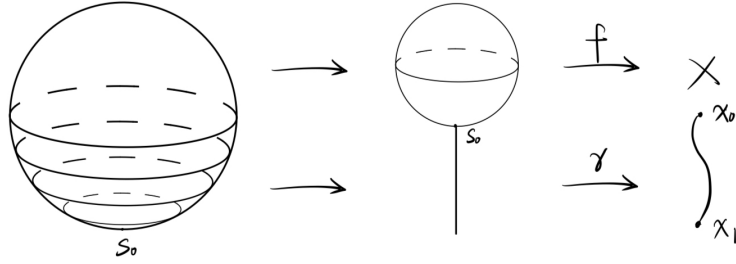
1.2.1 Dependency of base point

For a general space X , the homotopy group $\pi_n(X, x_0)$ is usually dependent on the base point x_0 . That is, given two points x_0 and x_1 , we usually can't find a well-defined isomorphism between $\pi_n(X, x_0)$ and $\pi_n(X, x_1)$.

Any path γ from x_0 to x_1 can induce a homomorphism

$$\gamma_* : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

The following figure provides a nice illustration.



One can check that γ_* is an isomorphism with inverse $\gamma_*^{-1} = \bar{\gamma}_*$. Moreover,

$$\gamma \simeq \gamma' \text{ rel } \{0, 1\} \implies \gamma_* = \gamma'_*.$$

Especially when the path γ starts and ends at the same point x_0 , that is, γ is a loop at x_0 , we get a $\pi_1(X, x_0)$ -action on $\pi_n(X, x_0)$ by:

$$\gamma \cdot \alpha = \gamma_*(\alpha), \quad \gamma \in \pi_1(X, x_0), \quad \alpha \in \pi_n(X, x_0).$$

Moreover, we have an equality

$$\pi_n(X, x_0) / \pi_1(X, x_0) = [S^n, X].$$

When $\pi_1(X, x_0)$ acts on $\pi_n(X, x_0)$ trivially, we say X is n -simple. Under this assumption, the isomorphism between $\pi_n(X, x_0)$ and $\pi_n(X, x_1)$ is canonically defined. Therefore we can use the notation $\pi_n(X)$ unambiguously.

定理 1.2.2. $\pi_n(X, x_0) / \pi_1(X, x_0) = [S^n, X]$.

证明. In order to avoid ambiguity, we use $[\alpha]_n$, $[[\alpha]_n]_1$, $[\alpha]$ to denote elements in $\pi_n(X, x_0)$, $\pi_n(X, x_0)/\pi_1(X, x_0)$, $[S^n, X]$ respectively.

The map between $\pi_n(X, x_0)/\pi_1(X, x_0)$ and $[S^n, X]$ is naturally given, that is,

$$\begin{aligned}\phi : \pi_n(X, x_0)/\pi_1(X, x_0) &\rightarrow [S^n, X] \\ [[\alpha]_n]_1 &\mapsto [\alpha].\end{aligned}$$

ϕ is surjective: For any $[\alpha] \in [S^n, X]$, let $x_1 = \alpha(s_0)$. If $x_1 = x_0$, then

$$\phi([[\alpha]_n]_1) = [\alpha].$$

If $x_1 \neq x_0$, there is a path γ from x_0 to x_1 , then

$$\phi([[\bar{\gamma}_*(\alpha)]_n]_1) = [\bar{\gamma}_*(\alpha)] = [\alpha].$$

ϕ is injective: If $[\alpha_0] = [\alpha_1]$, where $\alpha_0, \alpha_1 : (S^n, s_0) \rightarrow (X, x_0)$. By definition of $[S^n, X]$, there is a homotopy

$$H : S^n \times I \rightarrow X$$

satisfying $H(-, 0) = \alpha_0(-)$, $H(-, 1) = \alpha_1(-)$. We denote $\alpha_t(-) = H(-, t)$. Let

$$\gamma(t) := H(s_0, t), \quad t \in I.$$

This defines a loop at x_0 . Set

$$\gamma_\tau(t) = \gamma(\tau + (1 - \tau)t) = H(s_0, \tau + (1 - \tau)t), \quad t \in I.$$

γ_τ is a path from $\gamma(\tau)$ to $\gamma(1) = s_0$. Now we can give a homotopy between $\gamma_*(\alpha_0)$ and $(c_{x_0})_*(\alpha_1)$ which fixes the base point:

$$\begin{aligned}H' : (S^n, s_0) \times I &\rightarrow (X, x_0) \\ H'(-, t) &= (\gamma_t)_*(\alpha_t)(-).\end{aligned}$$

Since $\alpha_t(s_0) = H(s_0, t) = \gamma(t) = \gamma_t(0)$, the formula behind is well-defined. Moreover,

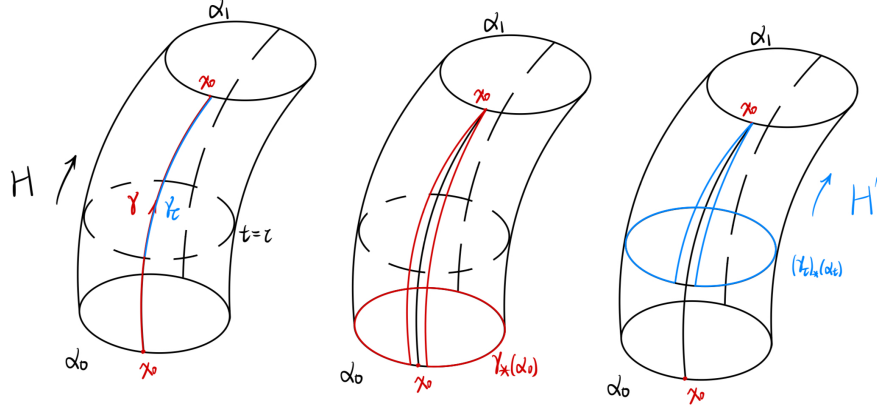
$$\begin{aligned}H'(-, 0) &= \gamma_*(\alpha_0)(-), \\ H'(-, 1) &= (c_{x_0})_*(\alpha_1)(-); \\ H'(s_0, t) &= (\gamma_t)_*(\alpha_t)(s_0) = x_0, \quad t \in I.\end{aligned}$$

So H' is what we need. Therefore

$$\gamma_*([\alpha_0]_n) = [(c_{x_0})_*(\alpha_1)]_n = [\alpha_1]_n.$$

So that

$$[[\alpha_0]_n]_1 = [[\alpha_1]_n]_1.$$



□

1.2.2 Homotopy Lifting Property & Homotopy Extending Property

命题 1.2.3 (Homotopy Lifting Property). *omit.*

引理 1.2.4 (Borsuk's Lemma). *Any CW pair (X, X') admits homotopy extending property.*

证明. Suppose X is a CW complex, X' is its subcomplex and $f' : X' \rightarrow Y$ can extend to $f : X \rightarrow Y$. For any homotopy $F' : X' \times I \rightarrow Y$ with initial map f' , we extend it by induction.

Step 1: For any vertex $x_0 \in X^0 - X'$, we define

$$F(x_0, t) = f(x_0), \quad t \in [0, 1]$$

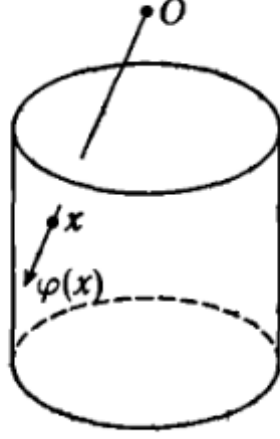
Step 2: Suppose we have extended the homotopy over the skeleton X^n . Now for any $(n+1)$ -cell

$$\psi : D^{n+1} \rightarrow X$$

The homotopy has been already defined on $S^n \times I$ and $D^{n+1} \times \{0\}$. To fill in the whole $D^{n+1} \times I$, we use the fact that there is a nice retraction (projection)

$$r : D^{n+1} \times I \rightarrow (S^n \times I) \cup (D^{n+1} \times \{0\})$$

The following figure provides a nice illustration.



Thus we can extend the homotopy to any $(n + 1)$ -cell, so as X^{n+1} . □

1.2.3 Exact Sequences for Fibrations

Given a fibration $p : E \rightarrow B$ with fibre F , we can derive an extremely useful long exact sequence of homotopy groups. First choose a base point $b_0 \in B$, then choose $e_0 \in p^{-1}(b_0)$. The fibering map p induces a homomorphism

$$p_* : \pi_n(E, e_0) \rightarrow \pi_n(B, b_0), \quad \forall n$$

Besides, the inclusion $i : F = p^{-1}(b_0) \hookrightarrow E$ induces

$$i_* : \pi_n(F, e_0) \rightarrow \pi_n(E, e_0), \quad \forall n$$

The key is to construct a connecting homomorphism ∂_* .

1.3 Whitehead Theorem

In CW complex category, there is a very useful theorem states that the weak homotopy equivalence is equivalent to homotopy equivalence. We will discuss this theorem in this section.

定理 1.3.1 (Whitehead Theorem). *A map $f : X \rightarrow Y$ between two CW complexes is a homotopy equivalence if and only if it induces isomorphisms on all homotopy groups.*

To proof this theorem, we start with a special case that Y is a point. The theorem states that a CW complex is contractible if and only if all its homotopy groups are trivial. That is,

定理 1.3.2 (Whitehead Theorem for contractible spaces). *A CW complex X is contractible if and only if $\pi_n(X, x_0) = 0$ for all $n \geq 0$ and $x_0 \in X^0$.*

证明. Let $f_0 = \text{id}_X$, $f'_0 = f_0|_{X^0}$ be the restriction of f_0 on the 0-skeleton X^0 . Since X^1 is path-connected, we can connect any $x_\alpha \in X^0$ to x_0 by a path $\gamma_\alpha : I \rightarrow X^1$. This defines a homotopy $H'_0(x, t) : X^0 \times I \rightarrow X$ between f'_0 and $X^0 \rightarrow x_0$. By Borsuk's lemma, H'_0 can be extended to a homotopy H_0 defined on the whole X . H_0 joints the identity map $H_0(x, 0) = f_0(x)$ to the map $H_0(x, 1) =: f_1(x)$ which sends X^0 to x_0 .

Let $f'_1 = f_1|_{X^1}$. We first construct a homotopy $H'_1(x, t) : X^1 \times I \rightarrow X$ between f'_1 and $X^1 \rightarrow x_0$ cell by cell: For any 1-cell e^1 , let $\psi : D^1 \rightarrow X$ be its characteristic map. The two edges of e^1 are vertexes $u = \psi(0)$ and $v = \psi(1)$. The homotopy H_0 takes u and v to x_0 , whose trajectories are $\gamma_u = H_0(u, -)$ and $\gamma_v = H_0(v, -)$ respectively. The homotopy between $f'_1|_{e^1}$ and constant map should be a map $H'_1 : D^1 \times I \rightarrow X$ such that $H'_1(x, 0) = \psi(x)$, $H'_1(x, 1) \equiv x_0$. Actually we can set

$$H'_1(u, t) = f_1(\gamma_u(t)), \quad H'_1(v, t) = f_1(\gamma_v(t))$$

Then $H'_1|_{\partial(e^1 \times I)}$ defines a loop at x_0 . Since $\pi_1(X, x_0) = 0$, $H'_1|_{\partial(e^1 \times I)}$ can extend to $e^1 \times I$. Notice that if $u \in e_\alpha^1 \cap e_\beta^1$ is the common vertex, then

$$H'_1|_{e_\alpha^1 \times I}(u, -) = H'_1|_{e_\beta^1 \times I}(u, -).$$

So we can merge them together to get H'_1 defined on $X^1 \times I$. Extending the homotopy to X , we get a homotopy H_1 joints f_1 to the map f_2 which sends X^1 to x_0 .

Similarly, using the equality $\pi_n(X, x_0) = 0$, we define a homotopy H_n between f_n and f_{n+1} such that f_{n+1} sends X^n to x_0 .

If the dimension of X is finite, we complete the proof. If $\dim X = \infty$, we combine all the homotopy H_n as follows. We first set that $H(x, t_n) = f_n$ where $t_n = \frac{2^n - 1}{2^n}$, then carefully fill in the gap by H_n . That is,

$$H(x, t) := H_n(x, 2^n t), \quad t \in [1 - \frac{1}{2^n}, 1 - \frac{1}{2^{n+1}}]$$

This homotopy joints id_X and the constant map. Since the restriction of H to $X^n \times I$ is continuous for all n , therefore the entire map H is continuous. $\square$

The construction of H'_1 in the proof can generalize to higher dimensions, which actually gives the following property:

命题 1.3.3. *If a CW complex X is n -connected, then its n -skeleton X^n is contractible in X .*

证明. First by the path-connectness of X , we can easily define $H_0 : X^0 \times I \rightarrow X$ which joints id_{X^0} and constant map.

For any 1-cell e^1 with characteristic map $\psi_{e^1} : D^1 \rightarrow X$, let v_0 and v_1 be its edges (they may be the same vertex). The trajectories of v_0 and v_1 under the homotopy H_0 are

$$\gamma_0 = H_0(v_0, -), \quad \gamma_1 = H_0(v_1, -).$$

The homotopy from e^1 to x_0 should be a map $H_1|_{e^1 \times I} : D^1 \times I \rightarrow X$ with top floor e^1 and bottom floor x_0 . To fill in the middle, we set the side to be γ_0 and γ_1 respectively, that is,

$$H_1|_{\partial(e^1 \times I)} = \begin{cases} \psi(x), & (x, t) \in D^1 \times \{0\}; \\ \gamma_i(t), & (x, t) \in \{i\} \times I, i = 0, 1; \\ x_0, & (x, t) \in D^1 \times \{1\}. \end{cases}$$

$H_1|_{\partial(D^1 \times I)}$ defines a loop at x_0 . Since $\pi_1(X, x_0) = 0$, H_1 can extend to $D^1 \times I$. Repeat the process to each 1-cell, notice that H_1 coincides at the common boundary of different cells, that is, if $v \in e_\alpha^1 \cap e_\beta^1$, then

$$H_1|_{e_\alpha^1 \times I}(v, -) = H_1|_{e_\beta^1 \times I}(v, -)$$

So we can merge all $H_1|_{e_\alpha^1 \times I}$ together to get $H_1 : X^1 \times I \rightarrow X$ such that $H_1(-, 0) = \text{id}_{X^1}$, $H_1(-, 1)$ is constant map.

Suppose we have defined $H_k : X^k \times I \rightarrow X$ between $H_k(-, 0) = \text{id}_{X^k}$ and $H_k(-, 1)$ the constant map. For each $(k+1)$ -cell e^{k+1} with characteristic map $\psi_{e^{k+1}}$, the map

$$H_{k+1}|_{\partial(e^{k+1} \times I)} = \begin{cases} \psi_{e^{k+1}}(x_0), & (x, t) \in D^{k+1} \times \{0\}; \\ H_k(x, t), & (x, t) \in \partial D^{k+1} \times I; \\ x_0, & (x, t) \in D^{k+1} \times \{1\}. \end{cases}$$

gives a $(k+1)$ -spheroid $\partial(D^{k+1} \times I) \rightarrow X$, which is contractible because of the triviality of $\pi_{k+1}(X, x_0)$. Thus $H_{k+1}|_{\partial(e^{k+1} \times I)}$ can extend to $D^{k+1} \times I$. Merge them gives the required map H_{k+1} . $\square$

There is a relative version of contractible case.

定理 1.3.4 (Relative Whitehead Theorem). *A CW pair (X, A) satisfies A being a deformation retract of X if and only if $\pi_n(X, A, x_0) = 0$ for all $n \geq 0$ and $x_0 \in A^0$.*

证明. ... $\square$

The Whitehead theorem can be derived from the relative version. The key is to use the mapping cylinder.